Windows PowerShell Common errors and reasons
Article Details

Problem Statement:

How can I troubleshoot PowerShell collection issues?

Version Affected:

All SL1 Stacks with PowerShell data collection

Diagnostic Steps:

SL1 can generate the following error messages when problems occurs due to Kerberos and Permission issues. For each error message, the reason and troubleshooting steps are listed below

Error Message	Reason	Troubleshooting Steps
Kerberos Issue		
Clock skew too great Clock skew too great while getting initial credentials	Time on collector or member server is different by more than 5 minutes (default setting)	1.Check time on the domain controller(KDC), collector, and domain devices to be monitored. 2.Setup NTP on Primary DC and configure devices to sync to it via Group Policy. 3.Make sure collector time syncing successfully by running "timedatectl" command on Collector.
Cannot find KDC for realm 'domain.com'	There is an issue in the /etc/krb5.conf file	1.On the collector open the /etc/krb5.conf file. 2.Make sure the domain to be contacted is in the [realms] section and matches the domain to be contacted (in upper case) 3.Make sure the kdc=dc.domain.com is pointed to the correct AD DC.
Server not found in kerberos database	IP address or hostname of the Windows server cannot be found on the KDC (domain controller)	1.Check that the device to be monitored is part of that domain used in the SL1 PowerShell credential, and the domain is correctly specified. 2.Check DNS and ensure that the hostname returned by DNS is identical to one the KDC contains. 3.If using hosts file entries, check that the correct IP is listed for the FQDN
Can't find client principal ‘name’ in cache collection	The kerberos keytab file was not found	1.This problem should resolve itself after a new collection as kinit is run again. 2.If not check the /tmp directory for the appropriate keytab file for the credential. Make sure the collector process can write to the location.
Configuration file does not specify default realm	Reverse DNS for KDC	1.On the collector open the /etc/krb5.conf file 2.For the specified KDC for the domain/realm follow the instructions at page bottom for checking forward and reverse lookup.
Decrypt integrity check failed	The password has been changed in AD	1.Update SL1 with the new password 2.If this is a child domain, add the domain and KDC for the child manually to the krb5.conf file in the REALMS section on the collector assigned to the device.

KDC has no support for encryption type	Your administrator probably restricted the encryption types to specific ones	Please check with your Windows Administrator which encryption types are supported for Kerberos.
		
		If you have some specific enctypes that they enforced, you may need to add that on collectors that monitor the devices that hit the issue. Open /etc/krb5.conf file , and add that into [libdefaults] section.
		
		Here is the example on how that looks like:
		`[libdefaults]`
		`default_tgs_enctypes = aes256-cts-hmac-sha1-96 des3-cbc-sha1 arcfour-hmac-md5 des-cbc-crc des-cbc-md5 des-cbc-md4 rc4-hmac`

Error Message	Reason	Troubleshooting Steps

		`default_tkt_enctypes = aes256-cts-hmac-sha1-96 des3-cbc-sha1 arcfour-hmac-md5 des-cbc-crc des-cbc-md5 des-cbc-md4 rc4-hmac`
		
		Here is a link that describes those options: https://web.mit.edu/kerberos/krb5-1.5/krb5-1.5/doc/krb5-admin/libdefaults.html#libdefaults
		
		To test if that works, you can use a command like below from the collector:
		`kinit powershell_user_used_for_monitoring@DOMAIN_IN_ALL_CAPS`
		
		Once you hit Enter, you will be prompted for a password for that user. If after you provide the password and hit Enter you get back to command prompt without any errors, you have added proper configuration.

Permission Issues		
get-wmiobject : Access denied	Monitoring user doesn't have access to WMI objects	1. Access the Management console. To do this: 2. In Windows Server 2008, click **Start**, right-click **Computer**, click **Manager**, and then expand **Configuration**. 3. In Windows Server 2016 and 2012, right-click the Windows icon, click **Computer Management**, and then expand **Services and Applications**. 4. Right-click on **WMI Control** and then select *Properties*. 5. On the **WMI Control Properties** window, click the **Security** tab, and then click the **Security** button. 6. Click the **Add** button, and then add the non-administrator user or group in the **Select Users, Service Accounts, or Groups** dialog, then click **OK**. 7. On the **Security for Root** window, select the user o group just added, then in the **Permissions** section at the bottom of the window, select the ***Allow*** checkbox for the *Execute Methods*, *Enable Account*, and *Remote Enable* permissions. 8. Under the **Permissions** section of the **Security for Root** window, click the **Advanced** button. 9. In the **Advanced Security Settings** window, double-click on the user account or group you are modifying. 10. On the **Permission Entry** window, in the ***Type*** field, select *Allow*. 11. In the ***Applies to*** field, select *This namespace and subnamespaces*. 12. Select the ***Execute Methods***, ***Enable Account***, and ***Remote Enable*** permission checkboxes, and then click **OK** several times to exit the windows opened for setting WMI permissions. 13. Restart the WMI Service from services.msc.
The specified credentials were rejected by the server	Variety of permissions issues	Solution-1: 1. Execute `winrm configSDDL default` on the Windows device and ensure the user has Read and Execute permissions. 2. Ensure the user is a local admin or a member of the following groups: 1. Performance Monitor Users 2. Event Log Readers 3. Distributed COM Users

Error Message	Reason	Troubleshooting Steps
		4. Remote Management Users 5. WinRMRemoteWMIUsers__ (if present on Windows device) 3. Make sure an AV or other security product is not blocking the user from connecting to WinRM. 4. If the HTTP/<FQDN> SPN is set on the server, try using WSMAN://%D or HOST://%D in the host field of the credential instead of %D. 5. This error sometimes gets bubbled up instead of the “clock skew too great”, check time and NTP settings. Solution-2: In the PowerShell credential being used, include the prefix **HOST** or **WSMAN** before the entry used in the Hostname/IP text field. For example: `HOST://%D` or `WSMAN://%D`. SL1 will use the WinRM service HOST or WSMan instead of HTTP and replace the variable with the IP address of the device that is currently using the credential.
Access is denied -Occurs only in last step of PS Credential Tester or after a successful pywinrm connection.	Execution of cmd.exe is blocked by GPO or other security product	1.Allow execution of cmd.exe for monitoring user
Credential		
Exception caught while communicating with http://SERVERNAME.DOMAIN:5985wsman (retrying connection)	Credential timeout is too low.	Increase credentials timeout that is used for the device.